

House Price Prediction — Summary of Findings

Internship Project — Week 1 | Name: [Arun singh] | Submitted: 21/05/2026

This report summarizes a regression analysis of 545 residential properties (Kaggle “Housing Prices Dataset”), built to predict sale price from features such as area, bedrooms, bathrooms, and amenities — and to identify which of those features matter most.

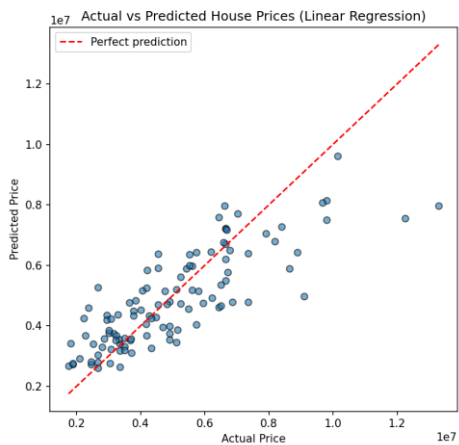
Which Features Influence Price the Most

Area is the single strongest driver of price, followed by the number of bathrooms and whether the home has air conditioning. Number of stories, parking space, and a preferred-area location also show a clear positive relationship with price, while unfurnished homes consistently sell for less than furnished ones.

How Accurate Is the Model

Model	MAE (₹)	RMSE (₹)	R ²
Linear Regression	9,70,043	13,24,507	0.65
Random Forest	10,22,560	14,01,497	0.61

In plain terms: the Linear Regression model explains about 65% of the price variation ($R^2 = 0.65$) and is typically off by roughly ₹9.7 lakh — about one-fifth of the average house price of ₹47.7 lakh. That is good enough for a ballpark estimate, but not a substitute for a professional appraisal.



Actual vs. predicted prices on the test set (Linear Regression)

What Surprised Us

Random Forest is often expected to beat Linear Regression, but here the simpler model edged it out (R^2 0.65 vs 0.61). With only 545 properties and price relating fairly linearly to features like area, Random Forest's extra flexibility didn't pay off and likely overfit the small training set slightly.

Recommendation for a Real Estate Business

Anchor pricing conversations around square footage and bathroom count first, since they are the most reliable price signals in this data. Treat amenities such as air conditioning and a furnished interior as relatively low-cost upgrades that meaningfully raise a property's perceived value.